# Question

Is there a statistically significant relationship between family size and household income after controlling for key demographic and employment variables?

# Background

- Family size plays a central role in how households allocate labor, resources, and income.
- Household income also depends on education, employment, marital status, race, and hours worked
- Understanding how family size affects household income is important for welfare policy, tax-credit design, poverty measurement, and resource allocation. This helps ensure that government programs reflect the economic realities of families of different sizes.

# Data

**Variable Codebook**
**Outcome Variable:** HHINCOME (total household income)
Total pre-tax household income from all sources earned by members of the household during the previous year
**Treatment Variable:** FAMSIZE (family size)
The total number of individuals living in the household
Demographics (age, sex, race, marital status), employment factors (employment status, class of worker, hours worked) and education.

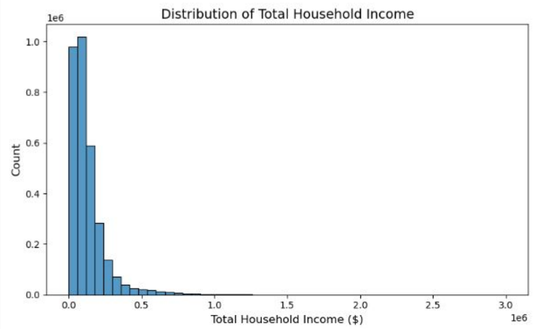
**Table 1: Descriptive statistics for continuous variables**

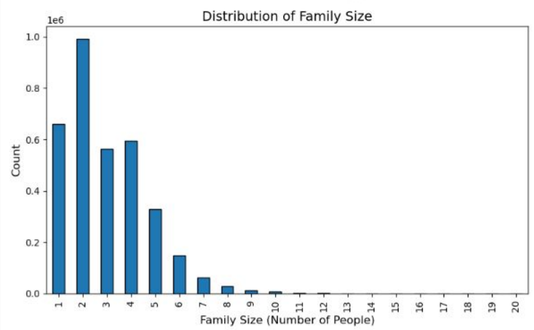
| | Total Household Income | Family Size | Age | Hours Worked / Week |
|---|---|---|---|---|
| N (sample size | 3228708.00 | 3405809.00 | 3405809.00 | 1758960.00 |
| Mean | 128725.74 | 3.00 | 43.11 | 37.69 |
| median | 95600.00 | 3.00 | 44.00 | 40.00 |
| Std. Dev. | 130647.68 | 1.72 | 23.97 | 13.16 |
| Minimum | 0.00 | 1.00 | 0.00 | 1.00 |
| Maximum | 3002000.00 | 20.00 | 96.00 | 99.00 |

**Table 2: Frequency Table for Categorical variables**

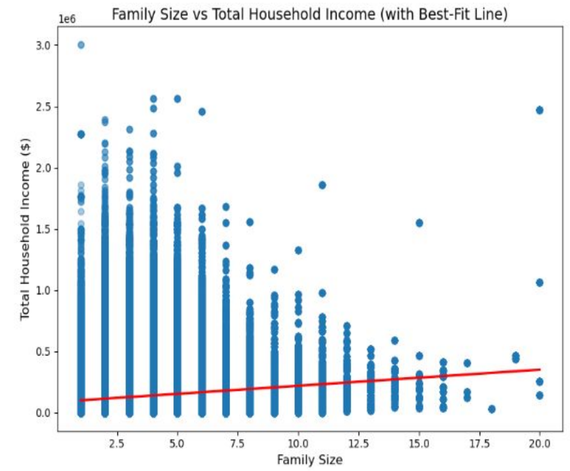
| Category | Percentage |
|---|---|
| sex | |
| - Male | 48.7% |
| - Female | 51.3% |
| Martial status | |
| - Married, spouse present | 49.2% |
| - Never married/single | 29.7% |
| - Divorced | 10.7% |
| - Widdowed | 6.6% |
| - Married, spouse absent | 2.3% |
| - Separated | 1.4% |
| Race (white/non-white) | |
| - White | 68.3% |
| - Non-White | 31.7% |
| Employment status | |
| - Employed | 56.1% |
| - Not employed | 43.9% |

# Histograms




# Scatter Plot




Slope (Coefficient for FAMSIZE): ≈ 15,000
Intercept: ≈ 13,700
This simple regression shows a positive relationship between family size and household income. However, the multiple regression model below provides a more accurate estimate of the effect of family size.

# Results

**coefficient for family size: β = –0.1179, p < 0.001**
(statistically significant)

- Family size is a statistically significant predictor of total household income ($p < 0.001$).
- The coefficient on family size indicates a correlated relationship between household size and income after controlling for age, education, employment status, marital status, race, hours worked, and year
- Although the effect of family size is statistically strong, the overall substantial power of the model is moderate
- Adjusted $R^2 \approx 0.167$
- The model explains only about 17% of the variation in a families household income, and the remaining percentage is determined by unobserved variables like regional income differences and job types
- Income is influenced by many unpredictable or unseen factors beyond what we can value as a number, see or determine meaning it's more than just family size that matters
- So basically Familt Size is a very important factor in a families annual income, it is a part of that control, not the result

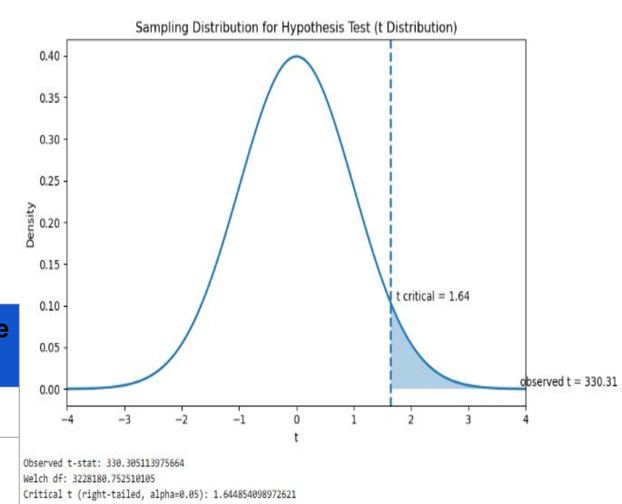
# Hypothesis Testing and Multiple Regression

**Hypothesis Testing:**
**Ho: Total household income of households with family size below the median is the same as or greater than the total household income of households with family size above the median.**

**Ha: Total household income of households with family size below the median is less than the total household income of households with family size above the median.**

**T-test of income and Family Size**

| | Low Family Size (< or = to 3) | High Family Size (> 3) |
|---|---|---|
| **Sample Size (N)** | **1,473,802** | **1,754,906** |
| **Mean Annual Income** | **10,3350.24** | **150036.56** |




**Since the t-statistic is greater than the critical value and the p-value is less than 0.05, the null hypothesis ($H_0$) is rejected.**

**Proposed Linear Model:**

**Total Household Income=α+β 1 (Family Size)+β2 (All Controls)+ε**

Multiple Regression Results (Poster Format)

| | Variable | Coef. (SE) | p-value |
|---|---|---|---|
| 0 | Intercept | 13684.87** (4469.62) | 0.002201 |
| 1 | C(SEX)[T.2.0] | -6029.92*** (803.20) | 6.081e-14 |
| 2 | C(RACE)[T.2.0] | -24467.59*** (1489.40) | 1.453e-60 |
| 3 | C(CLASSWKR)[T.2.0] | -4556.58*** (1269.09) | 0.0003303 |
| 4 | FAMSIZE | 15061.21*** (274.68) | 0 |
| 5 | AGE | 471.46*** (32.61) | 2.5e-47 |
| 6 | UHRSWORK | 959.34*** (32.03) | 2.91e-196 |
| 7 | N | 100000 | |
| 8 | Adj. R-squared | 0.167 | |

# Conclusion

There is a statistically significant relationship between family size and household income. After accounting for demographic and employment variables, larger households are associated with a higher income on average. But, because income does not increase proportionally with every additional household member, bigger families may naturally have more financial constraints per person, for example the number of adults and kids in a house. This means that while household income rises with the family size overall,m the economic well-being of families regardless of size depends on additional factors other than the total annual income alone

# References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